GRADUATE SCHOOL SOLAR ACADEMY

UNIVERSITÉ SAVOIE MONT BLANC

cnrs

Operational research for urban solar development

"Regional PV production modelling from open data"



Tuesday 20/10/2026 Session 2

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Curriculum

Plan

Day	Time	Duration	Content
Session 1 Monday 19/10	8h00 - 9h30	1 h 30	Lecture 70 % / Hands-on 30 %
Session 2 **Tuesday 20/10**	**13h15 - 14h45**	**1 h 30**	**Lecture 30 % / Hands-on 70 %**
Session 3 Wednesday 21/10 (TP 1) Thursday 22/10 (TP 2)	8h15 - 11h15 (TP 1) 13h00 - 16h00 (TP 2)	3 h	Lecture 15 % / Hands-on 60 % / Project 25 %
Session 4 Monday 09/11 (TP 1) Tuesday 10/11 (TP 2)	8h15 - 11h15 (TP 1) 8h15 - 11h15 (TP 2)	3 h	Lecture 10 % / Project 90 %
Session 5 Tuesday 24/11 (TP 2) Friday 27/11 (TP 1)	9h45 - 12h45 (TP 2) 13h15 - 16h15 (TP 1)	3 h	Project 100 %, then oral presentations
Total		12 h	

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Agenda

Review notebook 1

PV performance model steps 3 to 5

One hour through the whole chain

Notebook 2: Python intro 2, cell temperature and DC power

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Review notebook 1

Notebook recap

Google Colab link: https://colab.research.google.com/github/Mohamed-ENNHIRI/ENER958/blob/main/notebooks/python_intro_poa.ipynb

Correction: walked through in class

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Review notebook 1

Python commands 1/2

```
import numpy as np  # import the package numpy and rename it np
import pandas as pd  # import pandas, data structures and calculations
weather = pd.read_csv(path, index_col=0, parse_dates=True)  # read a csv with a datetime index
weather.head()  # show the first five rows
weather.shape  # number of rows and columns
weather["ghi"]  # select one column, a pd.Series
weather.loc["2023-06-21"]  # select one day with the datetime index
weather.loc["2023-06-19":"2023-06-25"]  # select a range of dates
weather["ghi"].resample("D").sum() / 1000  # daily energy in kWh/m2
weather["ghi"].plot(xlabel="time", ylabel="GHI, W/m2")  # plot with labelled axes
```

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Review notebook 1

Python commands 2/2

```
from pvlib.solarposition import get_solarposition  # import one function from pvlib
sp = get_solarposition(weather.index, lat, lon)  # zenith and azimuth for every time stamp
from pvlib.irradiance import get_total_irradiance
poa = get_total_irradiance(30, 180, sp["apparent_zenith"], sp["azimuth"],
                           weather["dni"], weather["ghi"], weather["dhi"], albedo=0.2)  # isotropic by default
poa["poa_global"].sum() / 1000  # annual POA energy, kWh/m2
def my_func(a, b=1):  # define a function, b is optional
    return a + 2 * b
my_func(1, b=3)  # returns 7
```

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One hour through the whole chain

Notebook 2: Python intro 2, cell temperature and DC power

Modeling steps

3. Shading, soiling, reflection

The quantity driving the current is the effective irradiance, not the POA irradiance.

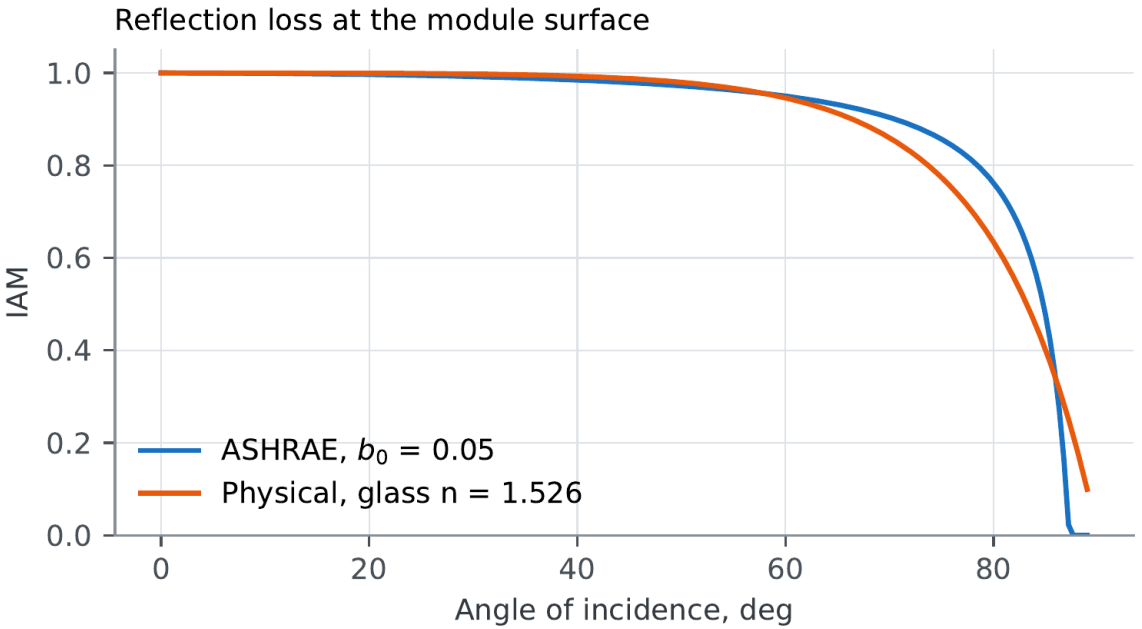
Reflection at the glass: the incidence angle modifier. Below 1 % near normal incidence, 5 % at 60°, 2 to 3 % on the annual energy.

Soiling: 1 to 3 % per year in a rainy climate.

Shading: site specific, unknown at fleet scale, one reason for a positive model bias.

In this course: spectral factor 1, soiling constant. Both go into the assumption list.

$$G_{eff} = G_{POA} \cdot \mathrm{IAM}(\theta) \cdot f_{spectral} \cdot (1 - L_{soil})(1 - L_{shade})$$



ASHRAE form, b0 = 0.05. King et al. (2004), SAND2004-3535. pvlib.iam.ashrae

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Modeling steps

4. Module and cell temperature

The hotter a module is, the less efficient it is.

Energy balance: absorbed radiation leaves as electricity or as heat. With α = 0.9 and η = 0.20, near 78 % of the absorbed energy leaves as heat.

Rooftop module at 900 W/m²: Tc − Ta reaches 25 to 35 K.

Wind removes heat. A ventilated open rack runs cooler than a roof-integrated module.

$$\alpha G_{eff} = \eta G_{eff} + U (T_c - T_a)$$

Modeling steps

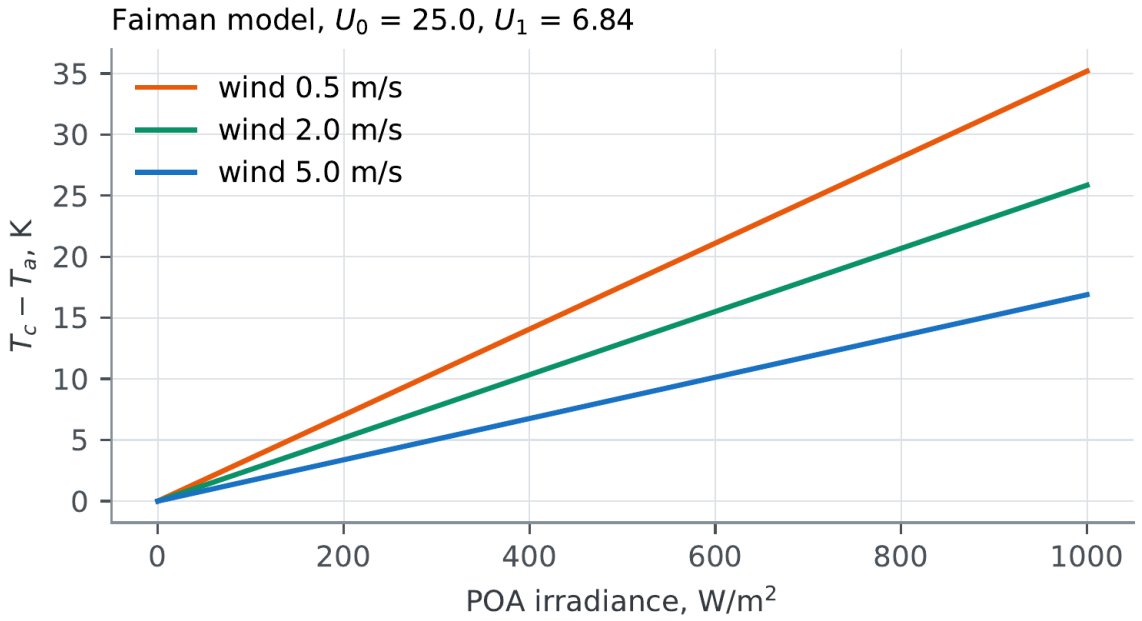
4. Cell temperature models

NOCT: Tc = Ta + (NOCT − 20) / 800 · GPOA, one parameter, no wind. NOCT = 45 °C.

SAPM: Tm = GPOA · exp(a + b · WS) + Ta, then Tc = Tm + GPOA / 1000 · ΔT. Open rack glass-polymer: a = −3.56, b = −0.075, ΔT = 3 K.

Faiman: Tc = Ta + GPOA / (U0 + U1 · WS). U0 = 25.0 W/m²K, U1 = 6.84 W/m³Ks. Used for the IEC 61853 energy rating.

Wind at 10 m over open terrain in the weather file, modules at 3 to 10 m behind obstacles: the least known input.



$$T_c = T_a + \frac{G_{POA}}{U_0 + U_1 v_w}$$

Faiman (2008). King et al. (2004). pvlib.temperature.faiman, pvlib.temperature.sapm_cell

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Modeling steps

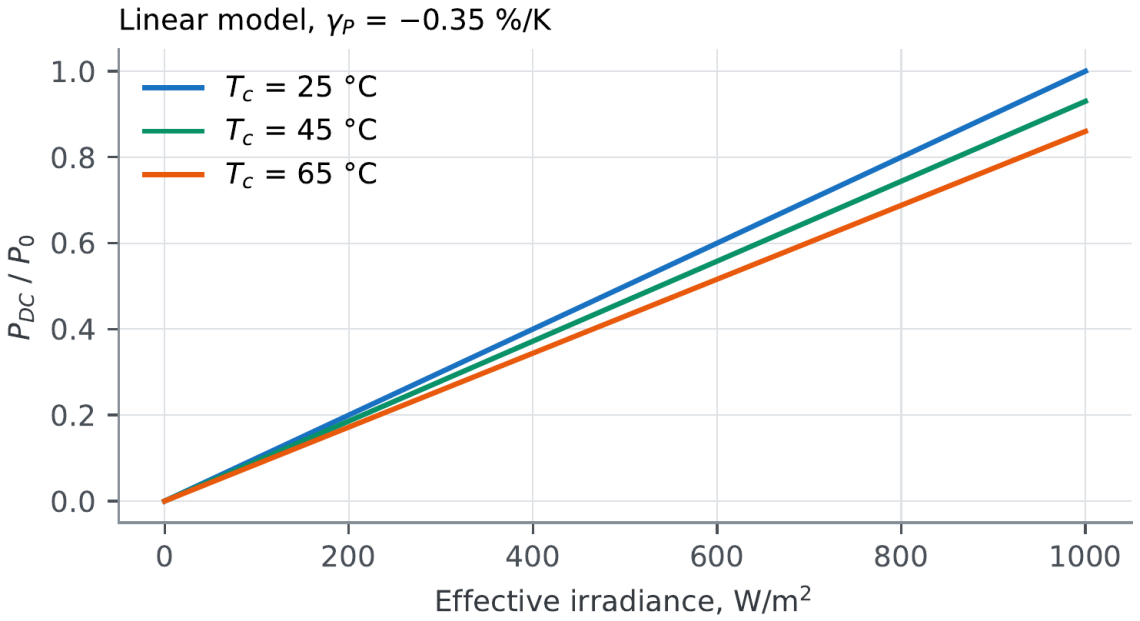
5. DC power

Linear efficiency model, the PVWatts form: nameplate P0 at STC (1000 W/m², 25 °C, AM1.5), temperature coefficient yP.

yP: −0.29 to −0.35 %/K for recent crystalline silicon, near −0.45 %/K for older modules.

Two parameters, both on every datasheet. Suited to a fleet of unknown modules.

Single-diode model: five parameters fitted per module type, reproduces the low-light behaviour. Not needed at fleet scale.



$$P_{DC} = P_0 \frac{G_{eff}}{1000} [1 + \gamma_P (T_c - 25)]$$

Evans (1981). Dobos (2014), PVWatts v5. De Soto et al. (2006). pvlib.pvsystem.pvwatts_dc

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One hour through the chain

Every number of steps 1 to 5, before you code it

Step	Model	Result
Inputs	Course site, 21 June 2023, 12:00 UTC. 30° south, 3 kWp	DNI 850, DHI 120 W/m², Ta 26 °C, wind 2 m/s
1. Solar geometry	NREL SPA, or the course formulas	zenith 22.64°, solar azimuth 193.3°
2. Angle of incidence	cos AOI from zenith, azimuth, tilt	AOI 9.38°
3. POA irradiance	Isotropic sky, albedo 0.2	direct 838.6 + diffuse 112.0 + reflected 12.1 = 962.7 W/m²
4. Cell temperature	Faiman, U0 = 25.0, U1 = 6.84	50.9 °C, 24.9 K above the air
5. DC power	PVWatts, γP = −0.35 %/K	2.626 kW, 9.1 % lost to temperature
6. AC power	Losses 14 %, inverter 0.96	2.168 kW
Performance	PAC / (P0 · GPOA / 1000)	instantaneous PR 0.75

Notebook 2 reproduces steps 4 and 5 in code. Your functions return the same numbers for the same inputs.

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Notebook 2

(Quick) Python introduction 2 and cell temperature / DC power

Google Colab link: https://colab.research.google.com/github/Mohamed-ENNHIRI/ENER958/blob/main/notebooks/python_intro2_temperature_dc.ipynb

First action: File, Save a copy in Drive.

I. Python introduction: filtering, if and loops, plots, functions.

II. The worked hour in code: three temperature models and the DC power, expected 50.9 °C and 2626 W.

Then one year: the spread between temperature models and the thermal loss.

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That’s it

Next session: losses, inverter, AC power, performance metrics. Then one system over a full year.